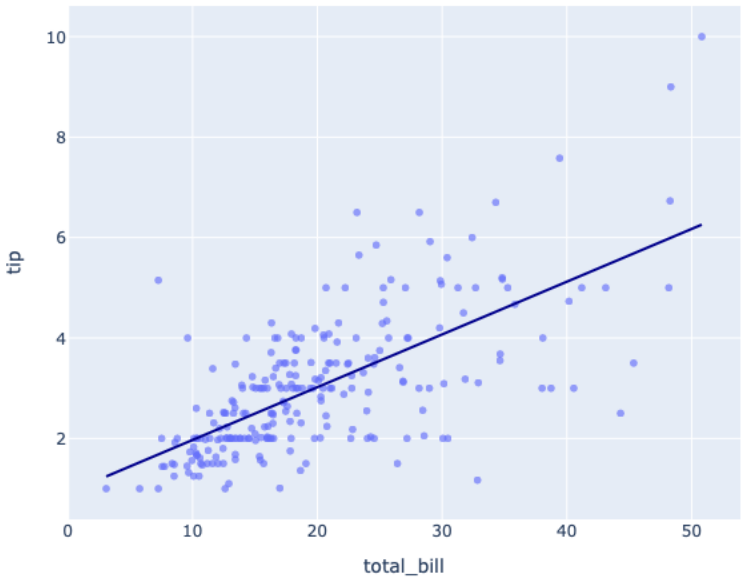
Activity Plotly

For this activity, I selected a graph related to machine learning regression in Python, specifically the “Ordinary Least Squares” implemented with the following code:

```
import plotly.express as px


df = px.data.tips()
fig = px.scatter(
    df, x='total_bill', y='tip', opacity=0.65,
    trendline='ols', trendline_color_override='darkblue'
)
fig.show()
```

This results in the following plot:



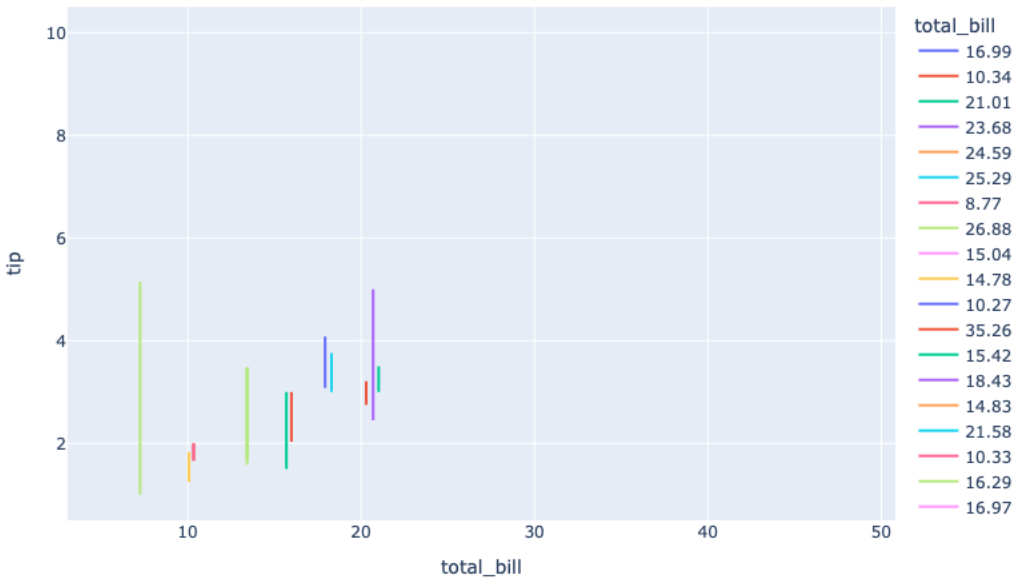
I then modified some aspects of the visualisation to observe their impact.

First, I created a line plot using the following code:

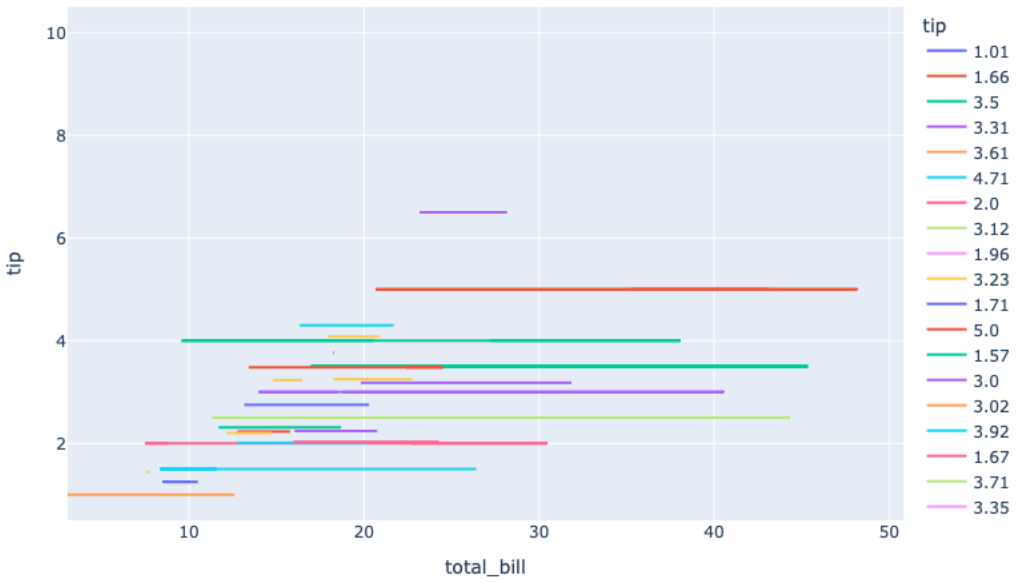
```
import plotly.express as px

df = px.data.tips()
fig = px.line(
    df, x='total_bill',
    y='tip',
    color = "total_bill"
)
fig.show()
```

This results in the following plot:



If we instead use “tip” as the color variable, the resulting plot is as follows:



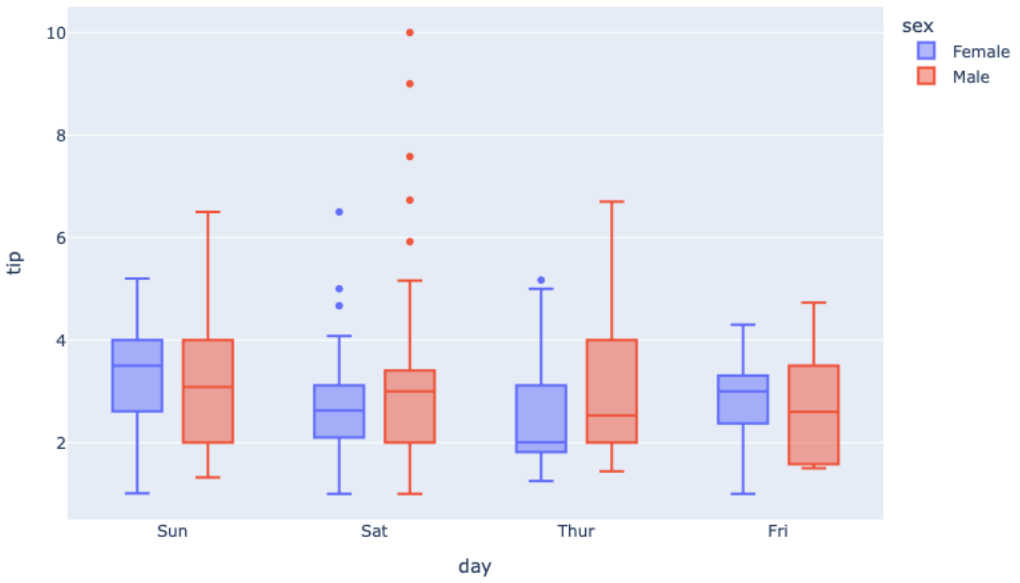
Considering both plots, the visualisation is difficult to interpret because the tip and total bill generate separate lines, with some values appearing only once.

I also created a box plot that shows the tips by day, divided by sex, using the following code:

```
import plotly.express as px
```

```
df = px.data.tips()
fig = px.box(
    df, x='day',
    y='tip',
    color = "sex"
)
fig.show()
```

This results in the following plot:



Overall, the scatter plot remains the most effective way to understand the relationship between tip and total_bill, as the line plot is less clear due to values that appear only once.

The box plot, on the other hand, highlights how tips vary across different days for both men and women.